Stormwater Regulation Summary: Philadelphia, PA

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Green City, Clean Waters — a $2.4B, 25-year green infrastructure initiative to reduce CSOs.
- PWD (Philadelphia Water Department) Stormwater Management Guidance Manual (v3.3) governs all development.
- Applies to ≥15,000 SF earth disturbance (5,000 SF in Darby/Cobbs Creek watershed) OR any new connection to the public stormwater system.
- First-inch retention from all Directly Connected Impervious Area (DCIA) is the core requirement.
- Parcel-based stormwater billing creates a direct financial incentive for every property owner to manage runoff.

2. Typical Soil Conditions

- Piedmont/Coastal Plain transition — soils range from Piedmont residual clays in NW to Coastal Plain sands in South/SW.
- Common soil series: Manor (stony loam), Chester (silt loam) in Piedmont; Sassafras (sandy loam) in Coastal Plain.
- Hydrologic Soil Group: Mixed — B in sandy Coastal Plain areas, C-D in Piedmont clay areas.
- Urban fill is prevalent in older developed areas, particularly Center City and along the Delaware/Schuylkill waterfronts.
- Implication: Infiltration is site-specific — feasible in sandy areas (South/SW Philly) but challenging on Piedmont clays. PWD accepts both infiltration and non-infiltration BMPs.

3. Green Roof Requirements

- Green roofs are not mandatory but are one of the most effective BMPs for meeting the first-inch DCIA requirement on constrained sites.
- PWD stormwater fee credits can offset up to 100% of the stormwater bill for projects that manage runoff on-site.
- Green roofs receive direct DCIA disconnection credit — the green roof area is removed from DCIA calculations.
- PWD Stormwater Management Guidance Manual provides specific design criteria and credit calculations for green roofs.
- Solarize Philly program provides group-buying discounts for solar — bio-solar systems address both stormwater and energy.

4. TSS Requirements

- PWD requires 80% TSS removal per the federal MS4 permit and CSO Long-Term Control Plan.
- Green roofs are approved BMPs with TSS removal credit — capturing the first inch inherently addresses the highest-TSS volume.
- PWD's Green City, Clean Waters plan ($2.4B over 25 years) relies heavily on green infrastructure (including green roofs) for pollutant load reduction.
- Additional treatment may be required for high-pollutant source areas (gas stations, industrial sites).
- Stormwater fee credit calculations factor in both volume reduction and pollutant removal from green BMPs.

5. Nature-Based Retention Requirements

- Retain the first 1.0" of rainfall from all Directly Connected Impervious Area (DCIA).
- For larger sites, PWD may require retention of up to 1.5" depending on downstream conditions.
- NRCS CN (Curve Number) method is the standard volume calculation approach.
- Typical post-development CN for impervious: 98; green roof-mitigated area: 65-75.
- PWD accepts green roofs, bioretention, permeable pavement, subsurface infiltration, and cisterns as retention BMPs.
- Channel Protection Volume (CPv): extended detention of the post-development 1-year, 24-hour storm (2.83") with drawdown in ≤72 hours.

6. Allowable Outflow Rates

- Channel Protection: 24-hour extended detention of 1-year storm — limits effective peak release.
- PWD does not specify a flat l/s/ha rate — peak flow control is relative to pre-development hydrology.
- For reference, typical pre-development urban release: 30-60 l/s/ha for the 10-year storm.
- *PWD reviews each project's drainage analysis individually — allowable rates depend on downstream sewer capacity.*
- Green roofs reduce peak discharge by attenuating rainfall and reducing total runoff volume — PWD quantifies this in their BMP calculations.